

Tables and graphs — build them for real

Participant worksheet. 90 minutes. Practice dataset first, then your own data.

Open **TB_Cohort_Practice_Dataset.xlsx**. Keep **Factsheet 5** (tables), **Factsheet 6** (charts) and **Factsheet 8** (Excel) beside you. Work in your group, but everybody builds it on their own laptop — you learn this with your hands, not by watching someone else's screen.

Part 1 — Warm-up on the practice data (40 min)

1 Clean first — find the six planted errors 10 min

Run the seven checks from Factsheet 4. There are six deliberate errors in this file. Find as many as you can and write them here. **Do not fix them yet.**

2 Build Table 1 12 min

Produce a baseline characteristics table for the cohort. Use a PivotTable (Factsheet 8, six steps). It must include:

- Age — decide first whether to report mean (SD) or median (IQR), and be able to say why
- Sex — n (%)
- HIV status — n (%), **including the unknowns as their own row**
- TB type — n (%)
- Distance to facility — median (IQR), *not* mean. You know why by now.

TIP

Before you build it, write your total N at the top of the table. Then check that every column adds up to it. If it does not, you have found either missing data or an error — both worth knowing about.

3 Build one chart, properly 10 min

Choose ONE of these and build it to publication standard:

- A histogram of *distance_to_facility_km* (bin width 5 km)
- A sorted bar chart of patients by municipality
- A bar chart of treatment success by municipality — y-axis starting at **zero**

Then run the five checks from Factsheet 6 on your own chart. Tick them off:

- ☐ Both axes labelled, with units
- ☐ y-axis starts at zero
- ☐ n is stated
- ☐ Survives black and white printing
- ☐ Decoration removed

4

Swap and critique 8 min

Swap laptops with the person next to you. Play Chart Doctor on *their* table and chart. Write one diagnosis and one prescription. Be kind, be specific, be honest.

Part 2 — Now your own project (50 min)

This is the part that matters

The practice dataset was the rehearsal. Now build the real thing. If your own data are not ready yet, build the **empty shells** of your tables and figures — all the row and column labels, titles and footnotes, with blanks where numbers will go. That is genuine progress, and your mentor can review it today.

5

Build your Table 1 20 min

Using the sketch you made this morning in Activity 7. Real numbers if you have them, empty shell if you do not.

6

Build your main results table or figure 20 min

The one that answers your research question.

7

Write the title and footnote for each 10 min

A title saying what, who, where, when, and N. A footnote explaining every exclusion. This takes five minutes and is the difference between a figure a reviewer trusts and one they query.

Before you close your laptop

- ☐ Is every percentage in my tables accompanied by its n and N?
- ☐ Does every table and figure have a title stating what, who, where, when and N?
- ☐ Does every exclusion have a footnote?
- ☐ Does every axis have a label with units?
- ☐ Does every bar chart y-axis start at zero?
- ☐ Have I saved my file with a name and date I will still understand in three months?
- ☐ Have I written down what my mentor told me to change?